



**ITCS442**

**Relational Database Design**

**Project Phase 2: Conceptual Database Design**

**Submitted by**

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**Presented to**

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## **Introduction**

**Project Title:** Restaurant Inventory Management System

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**Group ID:** 5

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**AI Tools Used:** No AI applied in this report

## User Requirement Specifications

### Changes from Phase 1

| Change Area                    | Previous Model (Phase 1)                                                                             | Updated Conceptual Model (Phase 2)                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Staff Management               | Staff were modeled as separate entities (e.g., Manager, Admin) despite sharing identical attributes. | All staff types are consolidated into a single 'Staff' entity with a 'position' attribute to distinguish roles, reducing data redundancy.                             |
| Menu – Ingredient Relationship | Ingredients were listed informally under each menu item.                                             | A formal many-to-many (M:N) relationship is defined between Menu and Ingredient using Menu_Ingredient, allowing flexible recipe definitions with required quantities. |
| Inventory Management           | Inventory tracking was not explicitly defined in the data model.                                     | Inventory is introduced as a separate entity linked to Ingredient to support stock monitoring, minimum stock levels, and inventory updates.                           |
| Supplier Information           | Supplier information was stored as unstructured text data.                                           | Supplier is modeled as a separate entity to support supplier management and ingredient sourcing.                                                                      |
| Procurement Process            | Ingredient purchasing and restocking processes were not supported.                                   | Purchase_Order and Purchase_Order_Detail are introduced to support ingredient procurement, cost tracking, and restocking operations.                                  |
| Payment Handling               | Payment information was not clearly represented in the conceptual model.                             | Payment is introduced as a separate entity with a one-to-one (1:1) relationship with Order to support payment tracking and status control.                            |

## Data Requirements

| User / Entity   | Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Staff           | <ul style="list-style-type: none"> <li>● staff_id (PK) – Unique identifier for each staff</li> <li>● citizen_id – National identification number</li> <li>● first_name – First name</li> <li>● last_name – Last name</li> <li>● position – Job title (e.g., Chef, Waiter, Cashier)</li> <li>● status – Employment status (e.g., Active, Inactive, Resigned)</li> <li>● gender – Male / Female / Other</li> <li>● phone – Contact phone number of the staff</li> <li>● created_at – Employment record creation date</li> </ul> |
| Order           | <ul style="list-style-type: none"> <li>● order_id (PK) – Unique identifier for each order</li> <li>● order_date – Date and time when the order is created</li> <li>● order_status – Current status of the order</li> <li>● total_price – Total price of the order</li> <li>● queue_number – Queue number used to identify the order sequence during order preparation</li> </ul>                                                                                                                                              |
| Menu            | <ul style="list-style-type: none"> <li>● menu_id (PK) – Unique identifier for each menu item</li> <li>● menu_name – Name of the dish</li> <li>● description – Brief description of the menu item</li> <li>● price – Selling price</li> <li>● category – Menu category (e.g., Appetizer, Main Course, Dessert)</li> <li>● status – Current availability status of the menu item used to control ordering</li> </ul>                                                                                                            |
| Payment         | <ul style="list-style-type: none"> <li>● payment_id (PK) – Unique identifier for each payment record</li> <li>● payment_method – Method of payment (e.g., Cash, Credit Card, E-Wallet)</li> <li>● paid_at – Date and time when the payment is made</li> <li>● amount – Total amount paid for the order</li> <li>● payment_status – Current payment status (e.g., Pending, Paid, Failed)</li> </ul>                                                                                                                            |
| Order_Menu      | <ul style="list-style-type: none"> <li>● quantity – Quantity of the menu item in the order</li> <li>● unit_price_at_time – Price of the menu item at the time of ordering</li> </ul>                                                                                                                                                                                                                                                                                                                                          |
| Ingredient      | <ul style="list-style-type: none"> <li>● ingredient_id (PK) – Unique identifier for each ingredient</li> <li>● ingredient_name – Name of the ingredient</li> <li>● unit – Unit of measurement (kg, liters, pieces, etc.)</li> <li>● price_per_unit – Cost per unit</li> </ul>                                                                                                                                                                                                                                                 |
| Menu_Ingredient | <ul style="list-style-type: none"> <li>● required_quantity – Quantity of the ingredient required for the menu item</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 |
| Supplier        | <ul style="list-style-type: none"> <li>● supplier_id (PK) – Unique identifier for each supplier</li> <li>● supplier_name – Name of the supplier</li> <li>● phone – Contact phone number of the supplier</li> <li>● address – Physical address of the supplier for contact and delivery purposes</li> </ul>                                                                                                                                                                                                                    |

|                       |                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inventory             | <ul style="list-style-type: none"> <li>● inventory_id (PK) – Unique identifier for each inventory record</li> <li>● quantity_available – Current stock quantity</li> <li>● last_updated – Date and time of last inventory update</li> <li>● minimum_stock_level – Minimum stock level before restocking</li> </ul>                                  |
| Purchase_Order        | <ul style="list-style-type: none"> <li>● purchase_id (PK) – Unique identifier for each purchase order</li> <li>● purchase_date – Date when the purchase order is created</li> <li>● total_cost – Total cost of the purchase order</li> <li>● purchase_status – Current status of the purchase order (e.g., Pending, Received, Cancelled)</li> </ul> |
| Purchase_Order_Detail | <ul style="list-style-type: none"> <li>● purchase_detail_id (PK)</li> <li>● quantity – Quantity of the ingredient ordered</li> <li>● unit_cost – Cost per unit of the ingredient</li> </ul>                                                                                                                                                         |

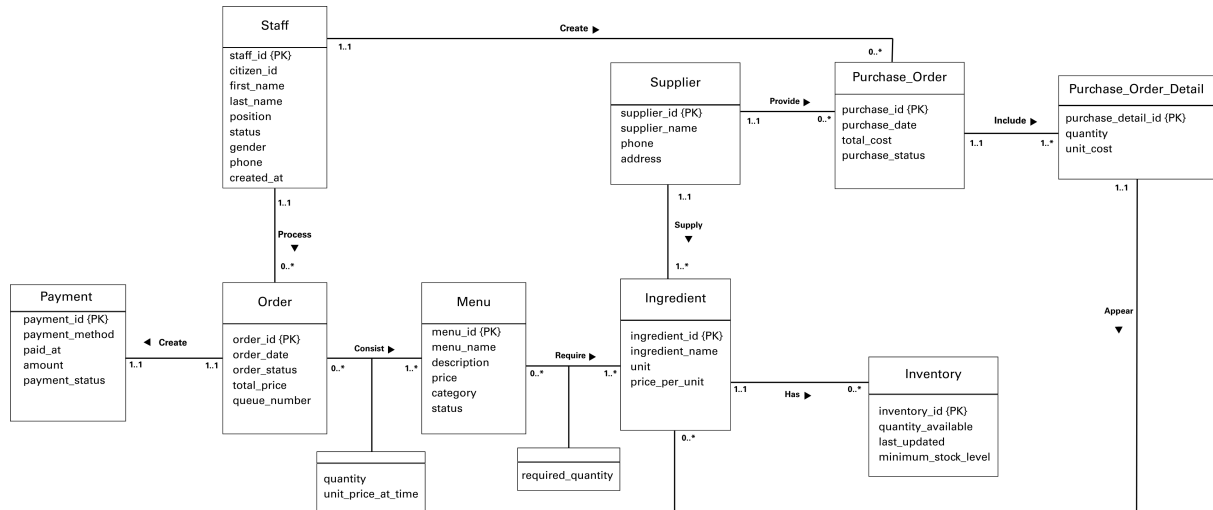
## Transaction Requirements

| User / Entity     | Data Entry                                                                               | Data Update / Deletion                                                                                                                                                                                 | Data Queries                                                                                                                                           |
|-------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Staff</b>      | <ul style="list-style-type: none"> <li>• Register a new order</li> </ul>                 | <ul style="list-style-type: none"> <li>• Update order status</li> <li>• Cancel a pending order</li> </ul>                                                                                              | <ul style="list-style-type: none"> <li>• View orders processed by staff</li> <li>• Find all pending orders</li> </ul>                                  |
| <b>Order</b>      | <ul style="list-style-type: none"> <li>• Create a new order record</li> </ul>            | <ul style="list-style-type: none"> <li>• Update order status (Confirmed, Completed)</li> <li>• Update order status to “Cancelled”</li> </ul>                                                           | <ul style="list-style-type: none"> <li>• Retrieve order details by order ID</li> <li>• List orders by date or status</li> </ul>                        |
| <b>Payment</b>    | <ul style="list-style-type: none"> <li>• Create a payment record for an order</li> </ul> | <ul style="list-style-type: none"> <li>• Update payment status</li> </ul>                                                                                                                              | <ul style="list-style-type: none"> <li>• Retrieve payment details by order ID</li> </ul>                                                               |
| <b>Menu</b>       | <ul style="list-style-type: none"> <li>• Add a new menu item</li> </ul>                  | <ul style="list-style-type: none"> <li>• Update menu price or category</li> <li>• Remove inactive menu items</li> </ul>                                                                                | <ul style="list-style-type: none"> <li>• List menu items by category</li> <li>• Find menu items below a specified price</li> </ul>                     |
| <b>Ingredient</b> | <ul style="list-style-type: none"> <li>• Add a new ingredient</li> </ul>                 | <ul style="list-style-type: none"> <li>• Update ingredient information</li> <li>• Remove expired ingredients</li> </ul>                                                                                | <ul style="list-style-type: none"> <li>• List ingredients by supplier</li> <li>• Retrieve ingredient details</li> </ul>                                |
| <b>Supplier</b>   | <ul style="list-style-type: none"> <li>• Register a new supplier</li> </ul>              | <ul style="list-style-type: none"> <li>• Update supplier information</li> <li>• Remove inactive suppliers</li> </ul>                                                                                   | <ul style="list-style-type: none"> <li>• List all suppliers</li> </ul>                                                                                 |
| <b>Inventory</b>  | <ul style="list-style-type: none"> <li>• Create initial inventory record</li> </ul>      | <ul style="list-style-type: none"> <li>• Update inventory quantity automatically after order confirmation</li> <li>• Update inventory quantity after purchase order is marked as “Received”</li> </ul> | <ul style="list-style-type: none"> <li>• View current inventory status</li> <li>• List ingredients with stock below the minimum stock level</li> </ul> |

|                              |                                                                                                                                |                                                                                                                                                              |                                                                                                                       |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Purchase_Order</b>        | <ul style="list-style-type: none"> <li>• Create a new purchase order</li> </ul>                                                | <ul style="list-style-type: none"> <li>• Mark purchase order as “Received” (restocking)</li> </ul>                                                           | <ul style="list-style-type: none"> <li>• Retrieve purchase orders by supplier or date</li> </ul>                      |
| <b>Purchase_Order_Detail</b> | <ul style="list-style-type: none"> <li>• Add ingredient items to a purchase order (ingredient, quantity, unit cost)</li> </ul> | <ul style="list-style-type: none"> <li>• Update ordered ingredient quantity or unit cost</li> <li>• Remove ingredient items from a purchase order</li> </ul> | <ul style="list-style-type: none"> <li>• Retrieve ingredient details included in a specific purchase order</li> </ul> |

# Conceptual Data Model

## ER Diagram



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# Entity Types, Keys, Attributes, and Constraints Summary

## 1. Staff (Strong Entity)

- **Purpose:** Stores information about staff members who work in the restaurant and process orders.
- **Primary Key:** staff\_id
- **Key attributes:** citizen\_id, first\_name, last\_name, position, status, gender, phone, created\_at

## 2. Order (Strong Entity)

- **Purpose:** Stores information about customer orders processed by staff members.
- **Primary Key:** order\_id
- **Key attributes:** order\_date, order\_status, total\_price, queue\_number

## 3. Payment (Strong Entity)

- **Purpose:** Stores payment information associated with each order.
- **Primary Key:** payment\_id
- **Key attributes:** payment\_method, paid\_at, amount, payment\_status

## 4. Menu (Strong Entity)

- **Purpose:** Stores information about menu items offered by the restaurant.
- **Primary Key:** menu\_id
- **Key attributes:** menu\_name, description, price, category, status

## 5. Ingredient (Strong Entity)

- **Purpose:** Stores information about ingredients used in menu items.
- **Primary Key:** ingredient\_id
- **Key attributes:** ingredient\_name, unit, price\_per\_unit

## 6. Supplier (Strong Entity)

- **Purpose:** Stores information about suppliers who provide ingredients to the restaurant.
- **Primary Key:** supplier\_id
- **Key attributes:** supplier\_name, phone, address

## 7. Inventory

- **Purpose:** Stores inventory information for tracking ingredient stock levels.
- **Primary Key:** inventory\_id
- **Key attributes:** quantity\_available, last\_updated, minimum\_stock\_level

## 8. Purchase\_Order (Strong Entity)

- **Purpose:** Stores information about purchase orders made to suppliers for ingredient restocking.
- **Primary Key:** purchase\_id
- **Key attributes:** purchase\_date, total\_cost, purchase\_status

## 9. Purchase\_Order\_Detail (Strong Entity)

- **Purpose:** Stores detailed information about ingredients included in each purchase order, including quantity and unit cost.
- **Primary Key:** purchase\_detail\_id
- **Key attributes:** quantity, unit\_cost

## 10. Order\_Detail (Associative Entity)

- **Purpose:** Represents the relationship between orders and menu items, storing the quantity ordered and the price at that time.
- **Primary Key:** (order\_id, menu\_id)
- **Key attributes:** quantity, unit\_price\_at\_time

## 11. Menu\_Recipe (Associative Entity)

- **Purpose:** Represents the relationship between menu items and ingredients, defining the quantity of each ingredient required for a menu.
- **Primary Key:** (menu\_id, ingredient\_id)
- **Key attributes:** required\_quantity

## Business Rules and Constraints

| Entities Involved                | Relationship        | Cardinality | Constraint | Description                                                      |
|----------------------------------|---------------------|-------------|------------|------------------------------------------------------------------|
| <b>Staff – Order</b>             | Process<br>[1:M]    | [1:M]       | (0,M)      | Each staff member may or may not process many orders.            |
|                                  |                     | [1:1]       | (1,1)      | Each order must be processed by exactly one staff member.        |
| <b>Order – Payment</b>           | Create<br>[1:1]     | [1:1]       | (1,1)      | Each order must create exactly one payment record.               |
|                                  |                     | [1:1]       | (1,1)      | Each payment record must belong to exactly one order.            |
| <b>Order – Menu</b>              | Consist of<br>[M:N] | [M:N]       | (1,M)      | Each order must consist of at least one menu item.               |
|                                  |                     | [M:N]       | (0,M)      | Each menu item may or may not appear in many orders.             |
| <b>Menu – Ingredient</b>         | Require<br>[M:N]    | [M:N]       | (1,M)      | Each menu item must require at least one ingredient.             |
|                                  |                     | [M:N]       | (0,M)      | Each ingredient may or may not be used in many menu items.       |
| <b>Supplier – Ingredient</b>     | Supply<br>[1:M]     | [1:M]       | (1,M)      | Each supplier can supply one or more ingredients.                |
|                                  |                     | [1:1]       | (1,1)      | Each ingredient must be supplied by exactly one supplier.        |
| <b>Staff – Purchase_Order</b>    | Create<br>[1:M]     | [1:M]       | (0,M)      | Each staff member may or may not create many purchase orders.    |
|                                  |                     | [1:1]       | (1,1)      | Each purchase order must be created by exactly one staff member. |
| <b>Supplier – Purchase_Order</b> | Provide<br>[1:M]    | [1:M]       | (0,M)      | Each supplier may or may not receive many purchase orders.       |
|                                  |                     | [1:1]       | (1,1)      | Each purchase order must belong to exactly one supplier.         |
| <b>Purchase_Order –</b>          | Include<br>[1:M]    | [1:M]       | (1,M)      | Each purchase order must include at least one detail line.       |

|                                           |                 |       |       |                                                                |
|-------------------------------------------|-----------------|-------|-------|----------------------------------------------------------------|
| <b>Purchase_Order_Detail</b>              |                 | [1:1] | (1,1) | Each detail line must belong to exactly one purchase order.    |
| <b>Purchase_Order_Detail – Ingredient</b> | Appear<br>[1:M] | [1:M] | (0,M) | Each ingredient may or may not appear in many purchase orders. |
|                                           |                 | [1:1] | (1,1) | Each detail line must refer to exactly one ingredient.         |
| <b>Ingredient – Inventory</b>             | Has<br>[1:M]    | [1:M] | (0,M) | Each ingredient may or may not have many inventory records.    |
|                                           |                 | [1:1] | (1,1) | Each inventory record must belong to exactly one ingredient.   |

## Pathway

### Data entry

Enter the details for a new staff member.

Enter the details for a new menu item (menu name, price, category, description).

Enter the details for a new ingredient (ingredient name, unit, price per unit).

Enter the details for a new supplier (supplier name, phone).

Enter the details for a new customer order (order date, order status, total price).

Enter the details of menus included in an order (quantity, unit price at time).

Enter payment details for an order (payment method, amount, paid at, payment status).

Enter the details for a new purchase order from a supplier (purchase date, status, total cost).

Enter the details of ingredients included in a purchase order (quantity, unit cost).

Enter inventory details when ingredients are stored (quantity available, last updated, minimum stock level).

### Data update/deletion

Update/delete staff details.

Update/delete menu details.

Update/delete ingredient details.

Update/delete supplier details.

Update/delete order details.

Update/delete order menu details (Order\_Detail).

Update/delete payment details.

Update/delete purchase order details.

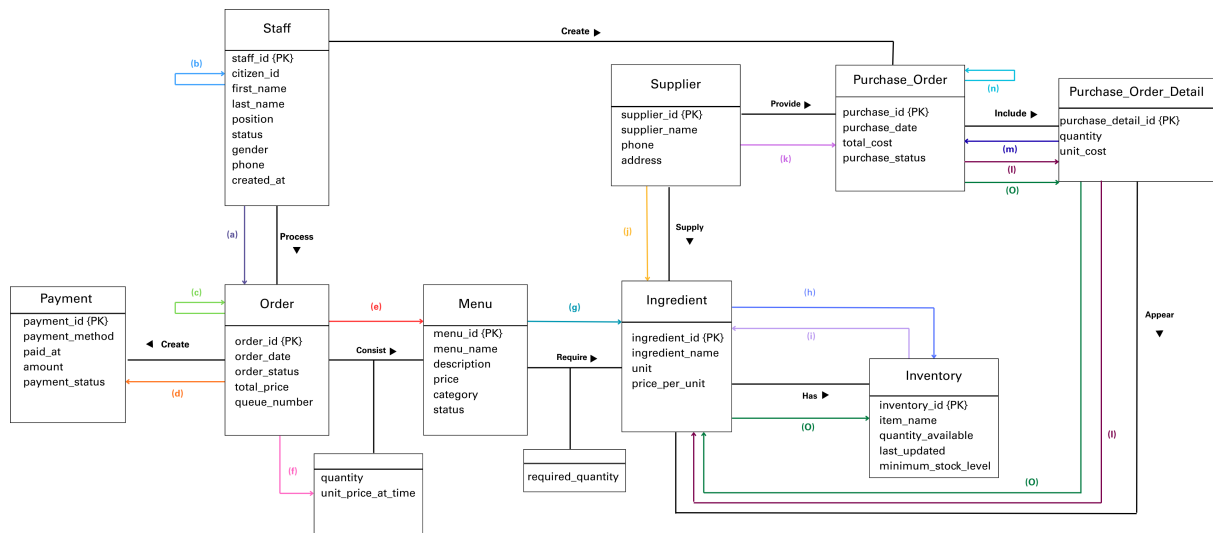
Update/delete purchase order ingredient details (Purchase\_Order\_Detail).

Update/delete inventory stock information.

### Data queries

Examples of queries required by the Staff user views:

- (a) List details of orders processed by a named staff member.
- (b) List all staff members alphabetically by their first name.
- (c) List all customer orders including order date, status, and total price.
- (d) Display the payment details of a given order.
- (e) List all menu items included in a given order, including quantity and unit price.
- (f) Calculate the total price (price\*quantity) of an order based on its order details.
- (g) List all ingredients required for a specific menu item.
- (h) Display current inventory stock level of each ingredient.
- (i) Identify ingredients with stock below the minimum stock level.
- (j) List details of all suppliers and the ingredients they supply.
- (k) List purchase orders provided by a given supplier.
- (l) Display ingredient details included in a given purchase order (quantity and unit cost).
- (m) List purchase orders that contain items with a unit cost higher than 300 baht.
- (n) Display all purchase orders with status Pending / Received / Cancelled.
- (o) Identify inventory items updated after receiving a purchase order.



[https://www.canva.com/design/DAHA7WP9fZ8/E4aVEEZNzRfjSmYf-BnEdg/edit?utm\\_content=DAHA7WP9fZ8&utm\\_campaign=designshare&utm\\_medium=link2&utm\\_source=sharebutton](https://www.canva.com/design/DAHA7WP9fZ8/E4aVEEZNzRfjSmYf-BnEdg/edit?utm_content=DAHA7WP9fZ8&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton)